

Keshav Bhagat

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EDUCATION

University of Michigan

May 2026

Bachelor of Science in Computer Science, Minor in Business (GPA: 3.84/4.0)

Ann Arbor, MI

- **Relevant Coursework:** Operating Systems; Distributed Systems; Database Management Systems; Web Systems; Intro to Artificial Intelligence; Data Structures and Algorithms; Theory of Computation; Computer Organization

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, Java, SQL, Go, FastAPI, Express, JavaScript, Dart, Flutter, React, Next.js

DevOps & Infra: PostgreSQL, MySQL, NoSQL, MongoDB, Firebase, Docker, Git, AWS, Prisma, SQLAlchemy, CI/CD

AI/ML: OpenAI API, Gemini API, LLM Tool Calling, Structured Outputs, Vector Search, pgvector

WORK EXPERIENCE

swype

June 2025 – Present

Co-Founder and Software Engineer

Ann Arbor, MI

- Built swype, an **AI-powered microlearning platform** with adaptive lessons and personalized quizzes driven by a 4-tier importance $\times$ 5-level familiarity matrix that adjusts difficulty and lesson sequencing to evolving user knowledge
- Designed structured AI content-generation pipelines with **schema-constrained outputs**, iterative prompting, and retrieval-based reuse, balancing **model quality, latency, and inference cost** to improve consistency, reduce hallucinations, and support scalable user experiences
- Architected stateless **FastAPI/PostgreSQL** backend with connection pooling and **Docker Compose** orchestration for AI-generated lessons, quizzes, and recommendations, designed for horizontal scalability ahead of production launch; secured with **Firebase Auth, JWT authorization**, and **CI/CD**-automated deployments
- Developed the iOS client in **Flutter** with **Google OAuth** and Provider-based state management; distributing to beta users via **TestFlight** ahead of production launch

Dream11

May 2024 – July 2024

Software Development Intern

Mumbai, India

- Centralized release coordination across engineering, QA, and release teams by building an internal release management platform (**Remix, Prisma, MySQL**), eliminating ad-hoc tracking for **800+ employees** across release creation, build history, and metadata
- Achieved **sub-100ms p99** core API latency by optimizing MySQL/Prisma queries through indexing and query rewrites, and by designing reusable schema patterns to support high-throughput internal release workflows
- Built dynamic UI workflows in **React/Remix** using Tailwind and reusable hooks, delivering forms, tables, filters, validations, and release-tracking components
- Wrote and executed **20+ performance tests** via **Callstack's Reassure** enforcing a **<5% render-time regression threshold** to catch performance drops in custom React Native components before release

Web Spiders

June 2023 – July 2023

Software Engineering Intern

San Jose, CA

- Built **EmpowerMeAI**, a job-seeking and career-advising chatbot tailored for job seekers, powered by the **GPT-3.5 Turbo**; ran a closed pilot, resulting in an **18% reduction in time-to-apply**
- Engineered the chatbot backbone with dynamic user-context injection, a career advisor system prompt, and per-session conversation history for multi-turn coherence

PROJECTS

PromptTube | *React, JavaScript, TypeScript, Manifest V3, Webpack*

May 2025 – Present

- Built PromptTube (Chrome Web Store Featured), an **AI-powered Chrome extension** with **600+ daily active users** and a **4.7-star rating** that enables **conversational querying**, multilingual Q&A, instant summaries, and **timestamp-grounded navigation** across long-form YouTube transcripts
- Designed a **conversational sidebar UX** injected into YouTube's DOM via MV3 content scripts and postMessage, persisting chat history, user preferences, and interaction state across sessions
- Engineered a **transcript retrieval pipeline** with resilient DOM extraction, timestamp alignment, retry handling, and caching strategies to support reliable conversational interactions over long-form video content

Sharded Key/Value Service with Paxos Replica Groups | *Go, Paxos, RPC*

November 2025

- Built a **sharded, fault-tolerant key/value store** with **linearizable** Get/Put/Append using **Paxos replication**
- Implemented a **Paxos-backed shardmaster** (Join/Leave/Move/Query) to rebalance shards and drive reconfigurations
- Added safe shard migration with **ErrWrongGroup** handling and **dedup state transfer** for at-most-once semantics
- Serialized shard movements through the Paxos log to preserve linearizability under concurrent reconfigurations